Regulation of gene expression in Prokaryotes

Genes are expressed only when needed

Regulation at:

- ✓ Transcription level (most **energy efficient**)
- ✓ Translation level

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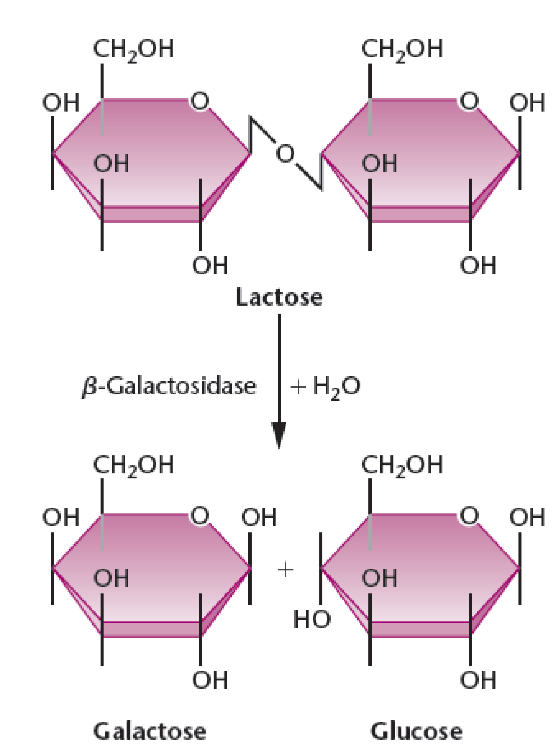
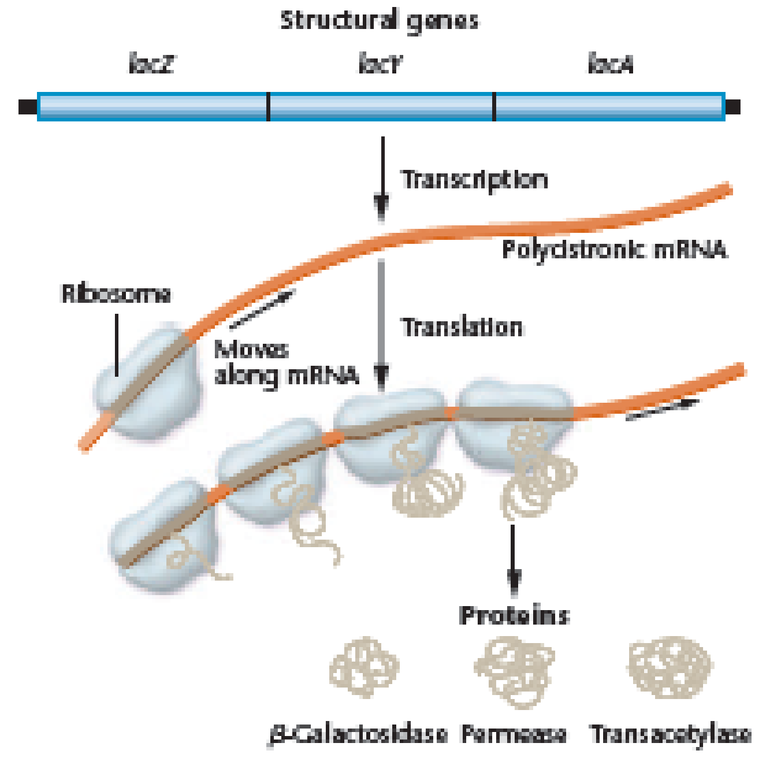
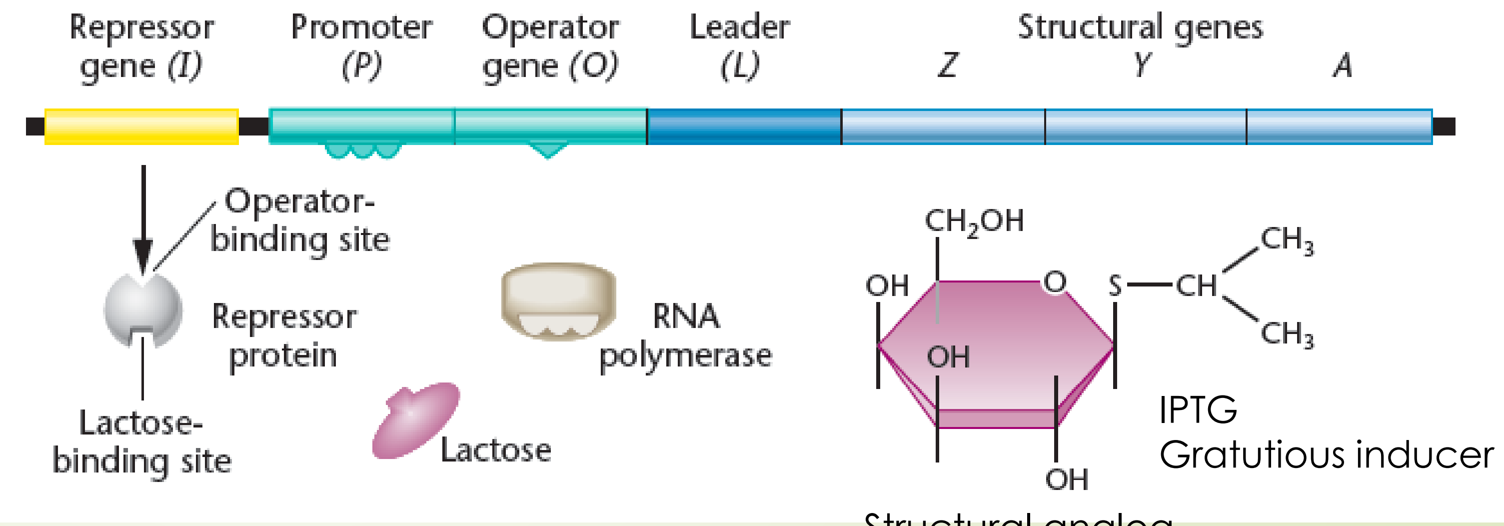
Lactose operon



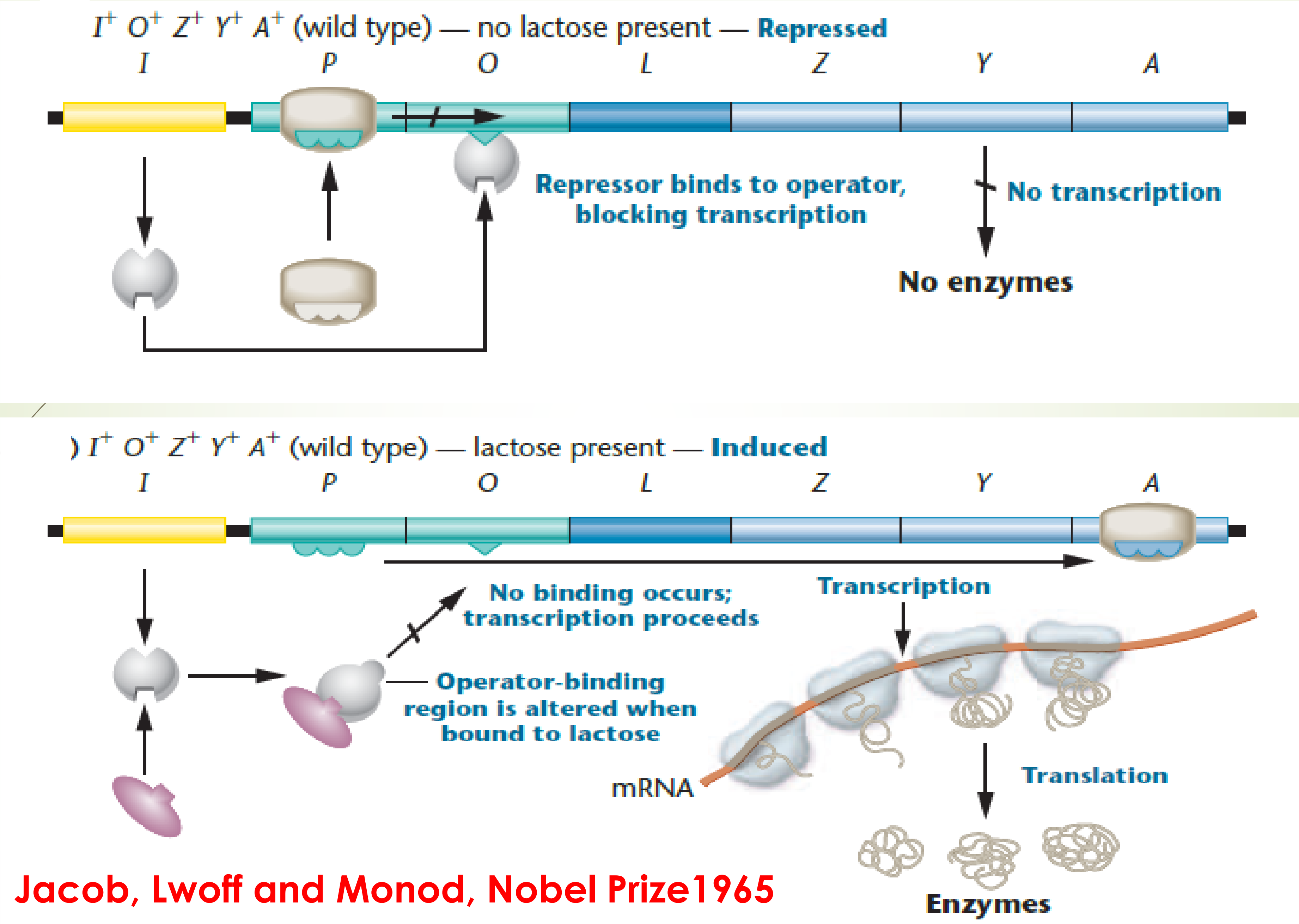
Expression of Lactose catabolism genes of *E. coli* are regulated to reduce metabolic burden





Structural analog

Model of Lac Operon (inducible system, negative control)



Jacob, Lwoff and Monod, Nobel Prize1965

Screening for *lacZ* expression

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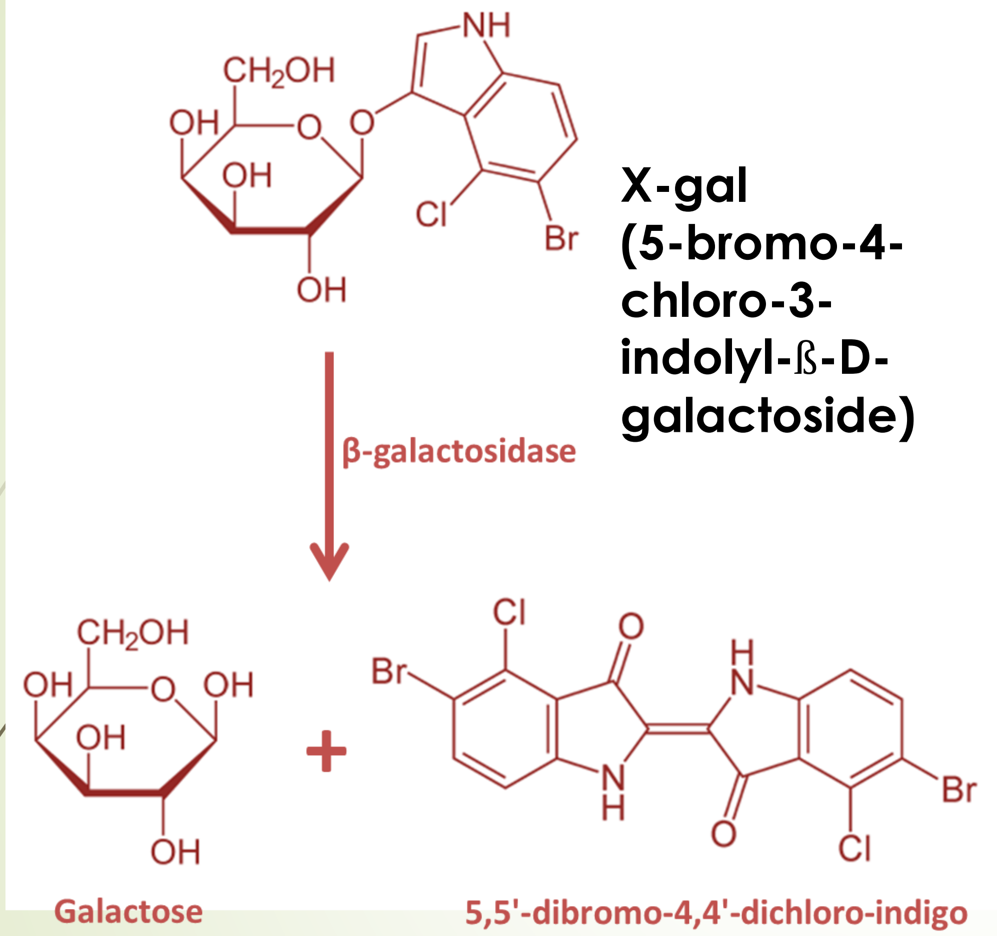
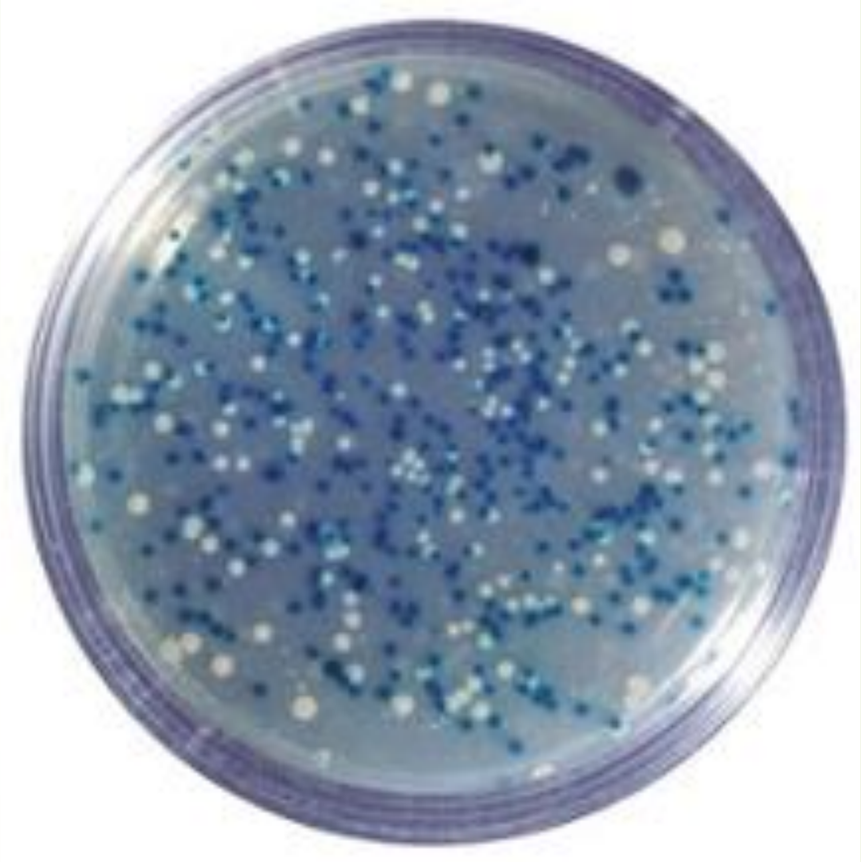


Blue color



Media contain X-gal + IPTG (inducer of lac operon)

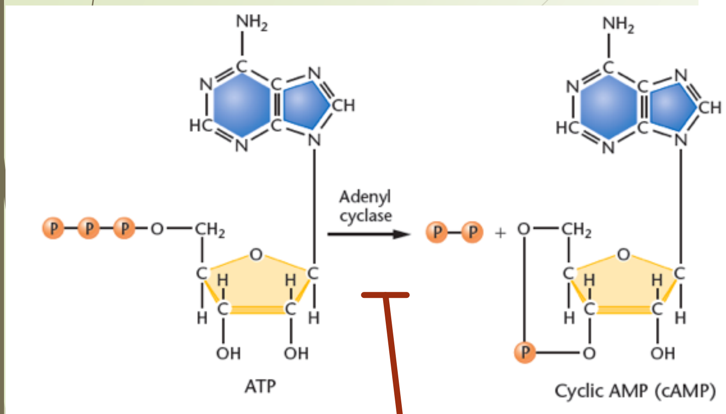
Other chromogenic substrates

ONPG → o-nitrophenol (yellow)

White colonies: ß-galactosidase not synthesized (lac^-)

Blue colonies: ß-galactosidase is synthesized (lac^+)

Catabolite Repression (Positive control)



Glucose (catabolite) repression